DSA03-Natural Language Processing

List of Lab Program

21. Create a python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

Code:

```
import nltk
import spacy
from nltk.corpus import wordnet as wn
nltk.download('wordnet', quiet=True)
nltk.download('omw-1.4', quiet=True)
nlp = spacy.load("en_core_web_sm")
sentence = "The intelligent student reads a book."
doc = nlp(sentence)
print("Sentence:", sentence)
print("\nNoun Phrases and Their Meanings:")
for chunk in doc.noun_chunks:
    print("Noun Phrase:", chunk.text)
    word = chunk.root.text
    synsets = wn.synsets(word, pos=wn.NOUN)
    if synsets:
        print("Meaning:", synsets[0].definition())
    else:
        print("Meaning: Not found in WordNet")
    print()
```

Output:

```
•••  Sentence: The intelligent student reads a book.

     Noun Phrases and Their Meanings:
     Noun Phrase: The intelligent student
     Meaning: a learner who is enrolled in an educational institution

     Noun Phrase: a book
     Meaning: a written work or composition that has been published (printed on pages bound together)
```